



ASSIGNMENT REPORT

Digital Design

Setup & Hold Timing Analysis (STA)

Course	Digital Design
Assignment	STA — Setup/Hold Analysis of a Two-Path Combinational Network
Student Name	Mohamed Hossam El-Sawy
Instructor / TA	Eng. Tamsah
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01 OBJECTIVE

This assignment performs a static timing analysis (STA) on a synchronous path in which two launching flip-flops (FF1, FF2) drive a shared combinational network — an inverter, an XNOR gate and a NAND gate — that feeds a common capturing flip-flop, at a clock period of 5 ns. The goal is to verify whether the path meets both the setup and hold requirements, to compute the resulting slack (or violation), and, if a violation is found, to select and place a buffer from the available library to correct it.

02 PROBLEM STATEMENT / REQUIREMENTS

Two flip-flops (Tc2q1, Tc2q2) launch data into a combinational block built from an inverter, an XNOR gate and a NAND gate, connected by the interconnect (wire) delays shown on the schematic. The NAND output feeds the D input of a third flip-flop with setup time T_{su} and hold time T_{hold} . Interconnect delay is included; clock uncertainty is not considered; both worst-case (max) and best-case (min) corners of the gate/flip-flop library are given.

- $T_{period} = 5$ ns; clock uncertainty is not considered; interconnect (wire) delays are included in every path.
- For each of setup and hold: determine whether timing is met, and calculate the slack (positive) or violation (negative) — using max-corner delays for the setup check and min-corner delays for the hold check.
- If a violation exists, select the smallest suitable buffer from the library (BUF0 / BUF1 / BUF2), state where it should be inserted, and confirm that the fix does not create a new setup violation.

03 DESIGN & ANALYSIS

3.1 Timing Paths in the Circuit

The inverter's output fans out in two directions, so there are three distinct combinational paths from a launching flip-flop to the D input of the capturing flip-flop:

- Path 1: FF1 $\rightarrow$ wire(0.2) $\rightarrow$ INV $\rightarrow$ wire(0.7) $\rightarrow$ NAND $\rightarrow$ wire(0.1) $\rightarrow$ D
- Path 2: FF2 $\rightarrow$ wire(0.3) $\rightarrow$ XNOR $\rightarrow$ wire(0.1) $\rightarrow$ NAND $\rightarrow$ wire(0.1) $\rightarrow$ D
- Path 3: FF1 $\rightarrow$ wire(0.2) $\rightarrow$ INV $\rightarrow$ XNOR $\rightarrow$ wire(0.1) $\rightarrow$ NAND $\rightarrow$ wire(0.1) $\rightarrow$ D

Setup is governed by the longest (worst-case, max-delay) path, since the last signal to arrive is the one that risks missing the setup window. Hold is governed by the shortest (best-case, min-delay) path, since the earliest signal to arrive is the one that risks violating hold.

3.2 Setup (Max-Delay) Analysis

Each path delay is built from $T_{c2q}(\text{max}) + \text{wire delays} + \text{gate delays}(\text{max})$. The setup check compares the longest path's total delay, plus $T_{su}(\text{max})$, against T_{period} .

Path	Tc2q (max)	Wire + Gate Delays (max)	Total Tpd (ns)
Path 1	0.4	$0.2 + 0.5 \text{ (INV)} + 0.7 + 1.0 \text{ (NAND)} + 0.1$	2.9
Path 2	0.5	$0.3 + 1.5 \text{ (XNOR)} + 0.1 + 1.0 \text{ (NAND)} + 0.1$	3.5
Path 3 (critical)	0.4	$0.2 + 0.5 \text{ (INV)} + 1.5 \text{ (XNOR)} + 0.1 + 1.0 \text{ (NAND)} + 0.1$	3.8

Path 3 is the setup-critical path (3.8 ns). Adding $T_{su}(\text{max}) = 0.8$ ns:

$$T_{pd}(\max) + T_{su}(\max) = 3.8 + 0.8 = 4.6 \text{ ns} \leq T_{\text{period}} = 5.0 \text{ ns}$$

Result: Setup timing is MET, with slack = $5.0 - 4.6 = +0.4 \text{ ns}$ (positive slack, no violation).

3.3 Hold (Min-Delay) Analysis

Each path delay is now built from $T_{c2q}(\min)$ + wire delays + gate delays (min). The hold check compares the shortest path's total delay against T_{hold} — using T_{hold} 's own worst-case (max) value, since that is the safe sign-off corner.

Path	T_{c2q} (min)	Wire + Gate Delays (min)	Total T_{pd} (ns)
Path 1 (critical)	0.3	0.2 + 0.3 (INV) + 0.7 + 0.8 (NAND) + 0.1	2.4
Path 2	0.4	0.3 + 1.2 (XNOR) + 0.1 + 0.8 (NAND) + 0.1	2.9
Path 3	0.3	0.2 + 0.3 (INV) + 1.2 (XNOR) + 0.1 + 0.8 (NAND) + 0.1	3.0

Path 1 is the hold-critical (shortest) path, $T_{pd}(\min) = 2.4 \text{ ns}$.

Real / sign-off analysis — uses $T_{hold}(\max) = 2.8 \text{ ns}$: slack = $2.4 - 2.8 = -0.4 \text{ ns} \rightarrow$ hold VIOLATION.

Naive / mis-corner tool analysis — uses $T_{hold}(\min) = 2.0 \text{ ns}$: slack = $2.4 - 2.0 = +0.4 \text{ ns} \rightarrow$ appears to pass, but this is not a valid sign-off corner.

Result: Using the correct (max T_{hold}) corner, Path 1 has a real hold violation of -0.4 ns and must be fixed.

3.4 Fixing the Hold Violation

Path 1 is too fast, so extra delay must be inserted in series on Path 1 only — specifically on the branch between the inverter's output and the NAND input, since that branch does not feed the XNOR gate and therefore will not add delay to Path 2 or Path 3. The minimum extra delay needed is 0.4 ns, measured on the buffer's min-delay rating (the hold check uses min-corner values).

Buffer	Min Delay (ns)	Max Delay (ns)	Resolves 0.4 ns Gap?
BUF0 (chosen)	0.4	0.6	Yes — exactly meets the 0.4 ns requirement
BUF1	0.8	1.1	Yes, but adds unnecessary extra delay
BUF2	1.3	1.4	Yes, oversized for this fix

BUF0 is the smallest buffer that closes the gap, so it is the correct choice — it fixes the violation while adding the least possible delay to the setup path.

Verification after inserting BUF0 on Path 1 only:

- Hold: $T_{pd1}(\min) + \text{BUF0}(\min) = 2.4 + 0.4 = 2.8 \text{ ns} = T_{hold}(\max) \rightarrow$ slack = 0 ns (violation resolved).
- Setup: $T_{pd1}(\max) + \text{BUF0}(\max) = 2.9 + 0.6 = 3.5 \text{ ns}$, still below the setup-critical Path 3 (3.8 ns) $\rightarrow$ setup slack stays +0.4 ns, unaffected.

Location: insert BUF0 in series on the inverter-output-to-NAND segment of Path 1, downstream of the fan-out point that feeds the XNOR gate — this confines the extra delay to the hold-violating path only.

04 RESULTS SUMMARY

Check	Critical Path	Path Delay (ns)	Requirement (ns)	Slack (ns)	Status
Setup	Path 3	3.8 (+0.8 Tsu)	Tperiod = 5.0	+0.4	MET
Hold (before fix)	Path 1	2.4	Thold = 2.8	-0.4	VIOLATED
Hold (after BUF0)	Path 1	2.8	Thold = 2.8	0.0	MET

05 DISCUSSION

The setup check passes comfortably because the setup-critical path (Path 3) is the longest electrical path — it goes through both the inverter and the XNOR before the NAND — leaving 0.4 ns of positive slack against the 5 ns period. The hold check fails because Path 1, the shortest path in the network, arrives at the capturing flip-flop before the required hold window closes when the flip-flop's worst-case hold time (2.8 ns) is used. This highlights a common STA pitfall: a hold check run against an overly optimistic Thold corner (2.0 ns here) would have hidden a real violation. Fixing the hold violation by inserting the smallest sufficient buffer (BUF0) on the shortest path only — rather than on a shared node — keeps the fix localized: it removes the hold violation entirely (0 ns slack) while leaving the setup-critical path, and therefore the setup slack, unchanged.

06 CONCLUSION

The two-launch, single-capture path meets setup timing with +0.4 ns of slack, but fails hold timing on its shortest path by 0.4 ns when analyzed at the correct (worst-case) hold corner. Inserting a single BUF0 buffer on the inverter-to-NAND segment of the shortest path resolves the hold violation exactly, without affecting the setup-critical path. The exercise illustrates the standard STA workflow: identify all combinational paths, use max-corner delays for setup and min-corner delays for hold, always sign off hold against the pessimistic Thold value, and localize any fix to the specific violating path.

07 APPENDIX — ORIGINAL HAND ANALYSIS

The scanned page below is the original hand-worked solution this report is based on. The setup path delays (2.9, 3.5, 3.8 ns), the hold path delays (2.4, 2.9, 3.0 ns), the Real-vs-Tool hold comparison, and the BUF0 fix all match the typed analysis and results in Section 3.

Assignment STA $T_{\text{Period}} = 5 \text{ ns}$

Setup Analysis:

$$T_{p1} = 0.4 + 0.2 + 0.5 + 0.7 + 1 + 0.7 = 2.9 \text{ ns}$$

$$T_{p2} = 0.5 + 0.3 + 1.5 + 0.1 + 1 + 0.1 = 3.5 \text{ ns}$$

$$T_{p3} = 0.4 + 0.2 + 0.5 + 1.5 + 0.1 + 1 + 0.1 = 3.8 \text{ ns}$$

Hold Analysis:

$$T_{p3} + T_{\text{su}} = 3.8 + 0.8 = 4.6 \text{ ns} < T_{\text{Period}} \rightarrow \text{No Violations}$$

Hold Analysis:

$$T_{p1} = 0.3 + 0.2 + 0.2 + 0.7 + 0.8 + 0.1 = 2.4 \text{ ns}$$

$$T_{p2} = 0.4 + 0.3 + 1.2 + 0.1 + 0.8 + 0.1 = 2.9 \text{ ns}$$

$$T_{p3} = 0.3 + 0.2 + 0.3 + 1.2 + 0.1 + 0.8 + 0.1 = 3 \text{ ns}$$

Real-Analysis $\rightarrow T_{\text{hold}} = 2.8 \text{ ns} \rightarrow \text{Negative slack, } t_{\text{slack}} = -0.4 \text{ ns}$

Tool-Analysis $\rightarrow T_{\text{hold}} = 2 \text{ ns} \rightarrow \text{Positive slack, } t_{\text{slack}} = 0.4 \text{ ns}$

③ If we assume Real Analysis Scenario:

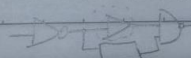
Buf0 $\rightarrow$ 

Figure A1: Original hand-written setup/hold timing analysis (scanned notes).